

LOGIKain

MASTER PLATFORM

FINAL PRODUCT REQUIREMENTS DOCUMENT

Company Website + CMS + SEO Engine + CRM + Client Portal + Project Operations + Billing + Automation + SaaS Foundation

Single source of truth for end-to-end AI-agent implementation

DOCUMENT STATUS

FINAL / implementation-ready. This document replaces version-based development roadmaps. Phase numbers describe build sequence only; all phases belong to one master product scope.

Think Clearly. Build Logically.

12 August 2026 | PRD Final 1.0

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Document Control

Authority, source basis, terminology and change policy.

Field	Final Specification
Product	LOGIKAiN Master Platform
Document	Final Product Requirements Document (PRD) + architecture + ERD + implementation plan
Version	Final 1.0
Date	12 August 2026
Primary audience	AI coding agent, developers, product owner, future technical team
Source basis	LOGIKAiN Company Profile Advanced v4 (2026) + V2 SEO-First specification + unified Phase 1-17 master roadmap
Primary stack	Next.js 16.x supported release + TypeScript + Supabase/PostgreSQL/Auth/Storage/RLS + Tailwind CSS; server-first public rendering
Deployment baseline	Vercel-compatible Next.js hosting + Supabase Cloud; provider interfaces for email/payment/analytics
Product direction	Service -> Partner -> Product -> Ecosystem
Roadmap model	One product scope. Phases are sequencing and dependency gates, not separate versions.

Source Alignment

- Core positioning: Technology, Creativity, Automation, Business Solutions; "Think Clearly. Build Logically."
- Six service groups: Software & Digital Product; AI, Automation & Integration; Branding & Digital Identity; Creative & Content; Document & Business Support; Digital Transformation.
- Operating process: Discover -> Analyze -> Design -> Build -> Test -> Launch -> Improve.
- Primary markets: UMKM, Startup, Personal Brand, School/Education, Organization/Community, Company/Corporate, Institution.
- Brand palette: Charcoal #171717, Copper #B36F43, Cream #F3F0EA; modern sans-serif typography.
- Long-term intent: grow from digital services and projects into long-term technology partnership, proprietary products and a technology ecosystem.

CHANGE POLICY

This PRD is the canonical implementation contract. The AI agent may make additive low-level engineering choices that preserve requirements. It must not remove modules, weaken security, change business workflows, or replace the data model with incompatible shortcuts without explicit product-owner instruction.

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01

Executive Product Blueprint

The final product is a business platform with a public acquisition layer, internal operating system and client-facing portal.

LOGIKain Master Platform is the official digital operating platform for LOGIKain. It begins with a high-performance corporate website and portfolio, but the final scope includes the internal tools required to acquire, convert, deliver, bill and support client work, plus a foundation for future proprietary SaaS products.

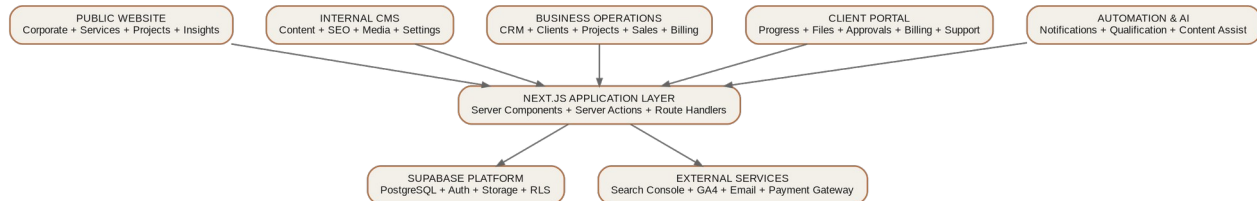


Figure 1. Final product architecture: acquisition, CMS, operations, portal, automation and shared platform services.

Platform Outcomes

Outcome	What the platform must enable
Acquire	Search-visible service, industry, case-study and insight pages that convert visitors into qualified requests.
Prove	Portfolio/case studies show problem, analysis, solution, implementation, technology and measurable outcomes.
Operate	Internal team manages content, leads, clients, projects, quotations, invoices, support and audit history from one system.
Collaborate	Clients can securely view progress, files, approvals, quotations, invoices and support conversations.
Automate	Repeatable notifications and business workflows reduce manual operational work.
Expand	Architecture supports proprietary SaaS products without forcing a rewrite of the corporate platform.

SEO EXPECTATION

The platform must maximize technical SEO readiness and organic-search opportunity, but no implementation may claim or guarantee a #1 search ranking. Ranking depends on competition, content quality, authority, links, intent alignment and search-engine systems outside the product.

Product Vision, Goals & Success Metrics

Business, operational, technical and search goals for the finished platform.

Vision

Create a single digital system that represents LOGIKaIn publicly, powers its internal delivery process, gives clients a professional self-service experience, and supports the company direction from service provider to technology ecosystem.

Primary Goals

- Build a premium, distinctive and trustworthy official company presence consistent with LOGIKaIn brand identity.
- Generate and qualify inbound business opportunities through SEO-led content and conversion paths.
- Enable routine website and portfolio updates entirely through CMS without source-code changes.
- Create a reusable sales-to-delivery data flow: Lead -> Quotation -> Client -> Project -> Invoice -> Support.
- Give clients a secure portal for visibility, files, approvals and commercial documents.
- Preserve security, auditability, accessibility, maintainability and performance as modules expand.
- Create extension points for automation, AI features and multi-tenant proprietary products.

Success Metrics

Area	Final success criterion	Target / Gate
Content operations	Admin can create, preview, publish, revise and restore core content without deployment.	PASS
Lead capture	Valid contact/start-project submission creates a lead and audit-safe event.	PASS
Sales lifecycle	Qualified lead can generate quotation and convert to client/project without duplicate re-entry.	PASS
Client delivery	Project milestones, tasks, files and approvals are usable by authorized internal users and portal members.	PASS
Billing	Accepted work can generate invoices and record payment status/history.	PASS
Security	Anonymous users cannot read drafts, leads, internal records, invoices or private client data.	PASS
SEO	Every indexable route has canonical metadata, sitemap eligibility and valid server-rendered primary content.	PASS
CWV	Target good field CWV: LCP <=2.5s, INP <=200ms, CLS <=0.1 at p75 where sufficient field data exists.	TARGET
Reliability	Critical writes are validated, logged and recoverable; migrations and backups exist.	PASS
Maintainability	Typed data access, module boundaries, tests and documentation are present.	PASS

03

Product Principles & Product Scope

Non-negotiable design principles and the final scope boundary.

Product Principles

Principle	Requirement
Logic before technology	Model the business problem and workflow first; use technology to support it, not to dictate it.
One source of truth	Public content, sales records, clients, projects and billing data must not require duplicate master records across modules.
Server-first public web	Indexable primary content must be present in server/static HTML; client JavaScript is added only for genuine interaction.
Structured, not chaotic	CMS uses structured content models and controlled blocks, not unrestricted drag-and-drop that can destroy design/SEO consistency.
Secure by default	Authorization is enforced in database policies and server code. UI hiding is never considered access control.
Audit important actions	Publishing, destructive changes, sales status changes, approvals, billing and privilege changes must be attributable.
Built to grow	New modules extend stable entities and contracts; schema changes use migrations.
Human-centered	Complex internal workflows must remain understandable to non-technical users.

Final In-Scope Product Domains

Domain	Included modules
Public Acquisition	Corporate website, services, industries, projects/case studies, insights, contact/start-project, legal pages.
Content Operations	CMS, media library, structured homepage/navigation, publishing workflow, revisions, activity logs.
Search Growth	Technical SEO engine, schema markup, internal linking, redirects, Search Console/GA integration hooks.
Sales	Lead CRM, qualification pipeline, quotation/proposal records, lead conversion.
Client Operations	Client/company records, contacts, relationships and history.
Delivery	Business projects, milestones, tasks, assignees, files, approvals, feedback.
Client Experience	Secure client portal, project visibility, files, approvals, quotes, invoices, support.
Commercial	Invoices, payments, retainers/subscription-ready billing abstractions.
Support	Tickets, messages, attachments, priority/status, resolution history.
Automation / AI	Notification jobs, workflow automation, optional AI-assisted capabilities with review and provider abstraction.
Product Expansion	SaaS/product tenant foundation, subscription-ready product entities and isolation patterns.

Deliberately Not Mandated

- A specific payment gateway, email vendor, AI provider or error-monitoring vendor is not hard-coded by this PRD; integrations must use provider adapters and environment configuration.
- Accounting/ERP, HR/payroll, inventory, tax filing and bank reconciliation are not required unless a future product decision adds them.
- AI-generated public content must never auto-publish without human review.
- Multi-language localization is not a launch prerequisite, but content models must not make later localization impossible.

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Users, Roles & Permission Model

Roles cover public visitors, internal operators and client portal members.

User Types

Role	Primary capabilities	Key restrictions
Visitor	Browse indexable content, filter projects, submit contact/start-project.	No private or draft data.
Editor	Create/edit content, upload media, manage drafts.	No user/role management, billing, destructive settings.
Sales	Manage leads, notes, quotations and client conversion.	No publishing or system administration by default.
Project Member	Manage assigned projects/tasks/files; participate in delivery.	No billing/admin unless separately granted.
Finance	Manage invoices/payment records and commercial reporting.	No content publishing or user privilege changes.
Support	Manage support tickets/messages.	No finance or publishing privileges.
Admin	Publish content, manage CRM/clients/projects/settings and most internal operations.	Cannot supersede owner-only security controls.
Owner	Full platform administration and role/security ownership.	Highest privilege; protected from accidental deletion.
Client Member	Portal access to assigned client/project data according to portal role.	Cannot access internal notes, other clients or admin/CMS.

Permission Rules

- Permissions must be capability-based in server code and RLS, even if roles provide default bundles.
- Client portal access requires an explicit membership relationship to a client account; authentication alone is insufficient.
- Finance documents are private by default and only visible to internal authorized users and explicitly authorized client members.
- Critical operations (role change, destructive deletion, invoice voiding, payment status override) require owner/admin privilege and audit logging.
- Service-role credentials are server-only and must never be exposed to browser bundles.

Information Architecture & Route Architecture

Public, internal and client-facing surfaces share one platform but have strict access and indexing boundaries.

Public Navigation Model

Primary route	Purpose
/	Brand proposition, service overview, proof, process, featured insights and conversion.
/about	Identity, philosophy, values, direction and team positioning.
/services	Service hub for six primary service groups.
/services/[slug]	Search-focused service landing page with related industries/projects/insights.
/industries	Cross-sector solution hub.
/industries/[slug]	Industry-specific needs, services, projects and insights.
/projects	Portfolio index with filters/search/pagination.
/projects/[slug]	Detailed case study.
/insights	Article/knowledge hub supporting authority and search clusters.
/insights/[slug]	Article detail with structured metadata and related content.
/process	Discover -> Analyze -> Design -> Build -> Test -> Launch -> Improve.
/contact	General inquiry.
/start-project	High-intent project request.

Protected Surfaces

```

/admin/*   Internal CMS + business operations. Authenticated. noindex.
/portal/*  Client self-service. Authenticated + client membership. noindex.
/api/webhooks/* Server-only provider callbacks. Signature-validated. Not user-browsable.

```

System Architecture & Final Tech Stack

Technology baseline optimized for SEO, structured business data, security and long-term extensibility.

Technology Baseline

Layer	Recommendation	Reason / Constraint
Framework	Next.js 16.x supported security release, App Router	Server/static rendering, metadata conventions, routing, full-stack server layer.
Language	TypeScript	Typed domain contracts across UI, server and data access.
UI	Tailwind CSS + custom LOGIKaIn design system	Fast implementation while preserving unique brand language.
Database	PostgreSQL via Supabase	Relational business/content data, JSONB selectively, full-text/search capabilities.
Auth	Supabase Auth	Integrated sessions with database-backed authorization.
Storage	Supabase Storage	Managed project/media/document assets with RLS-based access.
Authorization	Postgres RLS + server capability guards	Defense in depth.
Validation	Zod or equivalent shared schemas	Validate server inputs, forms and integration payloads.
Hosting	Vercel-compatible Next.js + Supabase Cloud	Independent application/data layers and straightforward deployment.
Search telemetry	Google Search Console + GA4 integration hooks	Organic visibility and conversion measurement.
Testing	Unit + integration + E2E + crawler/SEO smoke tests	Prevent regression across business and search-critical flows.

Architecture Rules

1. Public published content uses static/server rendering and caching/revalidation where safe; do not fetch primary SEO content only after hydration.
2. Privileged writes execute on trusted server code and remain subject to database RLS/policy rules.
3. Data access is grouped by feature/domain; raw database calls must not be scattered through presentational components.
4. Every schema change is a migration; production must never depend on ad-hoc dashboard edits.
5. Provider integrations (email, payment, AI) implement adapters so the provider can be replaced without rewriting business logic.
6. Public content tables and internal operational tables remain logically separated even when stored in the same PostgreSQL database.
7. Background/automation work must be idempotent and record status, attempts and failures.

Public Website Requirements

Public pages are the acquisition and trust layer of the platform.

ID	Capability	Acceptance intent	Priority
WEB-001	Home	Hero, positioning, service groups, problems solved, featured projects, process, industries, proof/testimonials, featured insights, primary CTA.	Must
WEB-002	About	Company story, identity/LOGIKAIN philosophy, values, direction and collective team positioning.	Must
WEB-003	Services Hub	All service groups with clear information architecture and links to detailed pages.	Must
WEB-004	Service Detail	Problem/solution framing, capabilities, suitable clients, process, related projects/industries/insights, CTA.	Must
WEB-005	Industries Hub	Market sectors and tailored digitalization opportunities.	Must
WEB-006	Industry Detail	Industry problems, relevant services, evidence, related insights, CTA.	Must
WEB-007	Projects Index	Search/filter/pagination; stable server URLs; featured items.	Must
WEB-008	Project Case Study	Structured proof with overview, problem, analysis, solution, implementation, technology, results, media and related content.	Must
WEB-009	Insights Index	Article index, categories, pagination and search-friendly URLs.	Must
WEB-010	Article Detail	Readable article, metadata, author/date, related content and structured data.	Must
WEB-011	Process	Explain Discover, Analyze, Design, Build, Test, Launch, Improve.	Must
WEB-012	Contact	General contact form with validation, anti-spam and lead creation.	Must
WEB-013	Start Project	Structured high-intent brief capturing service, budget, timeline and project context.	Must
WEB-014	Legal	Privacy and terms pages editable from controlled CMS content.	Must

Public UX Requirements

- Responsive from small mobile to desktop; no horizontal overflow.
- Primary navigation remains clear with services/projects/insights reachable in two interactions or fewer from home.
- Forms provide explicit validation, loading, success and safe error states; no silent failure.
- Animations are subtle, non-blocking and respect prefers-reduced-motion.
- Content is readable without JavaScript where feasible; interactive enhancement must not hide core content from crawlers or keyboard users.

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Portfolio & Case Study Engine

Completed work is reusable evidence, not a simple image gallery.

Project Content Model

Group	Required fields / relationships
Identity	title, slug, short_description, client display policy, project year, project type, status, featured, sort order.
Classification	industry, services many-to-many, technologies many-to-many, tags.
Narrative	overview, problem, analysis, solution, implementation, challenges, results, impact.
Evidence	metrics, gallery/media, mockups, optional video, external/live project URL.
SEO	seo_title, seo_description, og_image, canonical override only when justified, index flag.
Publishing	draft/review/published/archived, published_at, updated_at, author/reviewer.
Relationships	related projects, services, industry and insights generated or editorially curated.

Case Study Quality Gate

- Published case study must contain meaningful problem and solution content; placeholder-only projects are not indexable.
- Metrics must not be fabricated. If measurable results are unavailable, use qualitative outcomes and clearly label them.
- Project media requires descriptive alt text when meaningful; decorative images are explicitly marked decorative.
- Changing a published slug requires a permanent redirect from the old canonical path.

Insights / Content Engine

Insights create authority, internal-link depth and long-tail search coverage.

Content Types

Type	Primary purpose	Examples
Guide	Educational search demand	How to plan a company website; automation workflow guide.
Explainer	Clarify technology/business concepts	What is an internal dashboard; when a business needs API integration.
Case-driven article	Turn project learning into reusable insight	Lessons from digitizing manual operations.
Comparison	Help prospects evaluate options	Custom system vs template tool; website vs web app.
Industry insight	Support industry clusters	Digitalization for schools/UMKM/corporate operations.
Company update	Selective brand/news content	Launches, partnerships, major product releases.

Editorial Requirements

- Every article has explicit search/user intent, category, author, publication date, update date and related service/industry/project where relevant.
- CMS supports draft, review, scheduled/published and archived states; scheduled publication may be implemented through reliable job/revalidation mechanisms.
- Article routes remain stable; deleted/merged content must use redirect or appropriate 404/410 behavior based on content decision.
- AI assistance may suggest outlines, metadata or summaries, but human review is mandatory before publish.

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CMS & Publishing System

Internal users manage content, SEO, media and site configuration without code changes.

CMS Navigation

Dashboard
 Content: Projects | Services | Industries | Insights | Testimonials | Pages
 Growth: SEO | Redirects | Navigation | Homepage
 Business: Leads | Clients | Projects/Delivery | Quotations | Invoices | Support
 Assets: Media Library
 System: Users | Roles | Activity | Automation | Settings

Core CMS Requirements

ID	Requirement	Priority
CMS-001	Protected admin layout with role/capability-aware navigation.	Must
CMS-002	CRUD + preview + workflow for projects, services, industries, posts and testimonials.	Must
CMS-003	Media library with upload, alt text, metadata, reuse and access policy.	Must
CMS-004	SEO panel with SERP preview, metadata, index rules and technical checklist.	Must
CMS-005	Navigation/homepage structured section controls.	Must
CMS-006	Revision snapshots for high-value content and restore workflow.	Must
CMS-007	Activity log for important mutations and privileged actions.	Must
CMS-008	Search, filters, pagination and bulk-safe operations on large collections.	Must
CMS-009	Publish/revalidate behavior must update public routes without full redeploy.	Must
CMS-010	Destructive operations require clear confirmation and preserve audit trail.	Must

Publishing State

```
draft -> review -> published -> archived
      ^           |
      |-----| (return for edits when needed)
```

SEO-First Architecture

Technical SEO is a platform primitive, not a final pre-launch checklist.

Indexability Rules

Surface	Index policy	Rendering / crawl behavior
Public canonical pages	index, follow when published and useful	Static/server rendered primary content; canonical self-reference unless intentionally canonicalized.
Draft/preview	noindex; authenticated or signed preview	Never present in sitemap.
Admin	noindex + authentication	Do not depend solely on robots.txt for privacy.
Client portal	noindex + authentication + membership authorization	Private data never relies on crawler directives as security.
Internal search/filter combinations	Canonical to stable index page or noindex when combinations create thin/duplicate URLs	Avoid uncontrolled faceted index explosion.
Old slug	301/308 permanent redirect to current canonical route	Redirect chain must remain one hop where possible.

SEO Engine Requirements

- Dynamic metadata for Project, Service, Industry, Post and standard pages with safe generated fallback.
- Per-entity SEO title, meta description and OG image; canonical URL resolver; index/follow controls.
- Dynamic sitemap containing only canonical, public, indexable URLs; split sitemaps when scale requires.
- robots.ts/robots.txt for crawl guidance; private data remains protected by authentication/authorization, not robots rules.
- JSON-LD helpers for Organization, WebSite, WebPage, Service, Article, BreadcrumbList and other schema only when page content supports it.
- Breadcrumbs visible to users where appropriate and consistent with structured data.
- Internal-link relations connect services <-> industries <-> projects <-> insights.
- Redirect manager stores old path, destination, status and creation reason; published slug changes create redirect transactionally.
- SERP preview and pre-publish SEO checklist in CMS. Checklist is quality control, not a ranking guarantee.
- Crawler smoke tests assert status code, canonical, robots meta, title, description, H1 presence and sitemap membership.

Performance / Core Web Vitals Targets

Metric	Good target	Measurement rule
LCP	<= 2.5 seconds	75th percentile field data where sufficient traffic exists; lab used pre-launch only as regression signal.
INP	<= 200 ms	75th percentile field data.
CLS	<= 0.1	75th percentile field data.

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Lead CRM

Inbound requests become structured opportunities with ownership and a traceable pipeline.

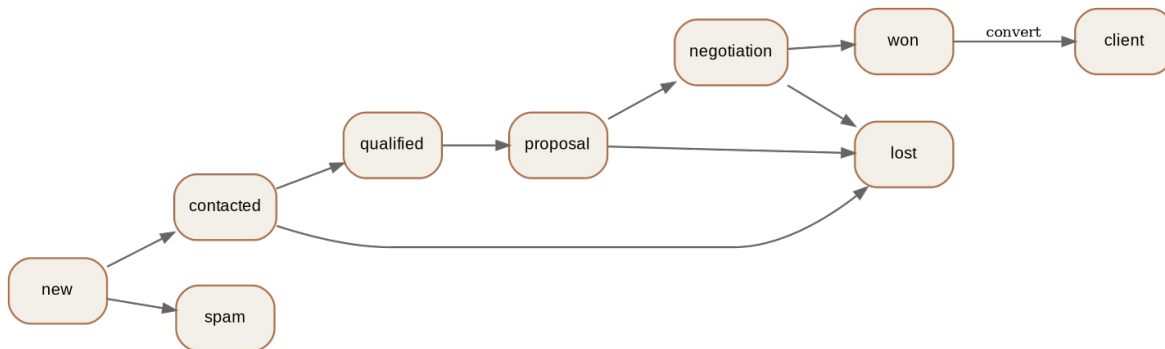


Figure 2. Default lead lifecycle.

Lead Requirements

Capability	Requirement
Capture	Contact and start-project forms create leads through validated server operations; store source, UTM/referrer when available and consent context where applicable.
Deduplication	Warn/associate on matching email/phone rather than silently creating uncontrolled duplicates.
Pipeline	new, contacted, qualified, proposal, negotiation, won, lost, spam.
Notes	Internal notes with author and timestamp; never exposed to client portal.
Assignment	Assign owner/sales user; provide filter/search.
History	Status changes and important edits create timeline/audit events.
Conversion	Won/qualified lead can create/link client, contact, quotation and project without retyping core contact data.

Lead Detail

- Contact identity and company context.
- Service interest, budget range, timeline and project description.
- Source and campaign/referrer context when legally/technically available.
- Status, assignee, notes, quotations and linked client/project.
- Activity timeline.

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Client Management

A client account is the long-lived business entity after or alongside a lead.

Client Model

Entity	Purpose
clients	Company/person master record used by projects, quotes, invoices, portal membership and support.
client_contacts	One client may have multiple contacts with roles and preferred communication details.
client_members	Maps authenticated portal users to a client account with portal role/capabilities.
client_notes	Internal relationship notes, separate from client-visible project feedback.
client_documents	Commercial/general client documents not tied to a single project when needed.

Client Requirements

- Client record must not be automatically exposed to a portal user; explicit membership is required.
- Lead conversion should preserve source history and link the resulting client back to originating lead(s).
- Client status supports active, inactive, prospect/converted context as appropriate; hard deletion of commercially referenced clients is forbidden.
- Client timeline may aggregate quotations, projects, invoices and support without duplicating those records.

Project Management & Delivery Operations

Internal delivery mirrors LOGIKAiN process while staying lightweight enough for real work.

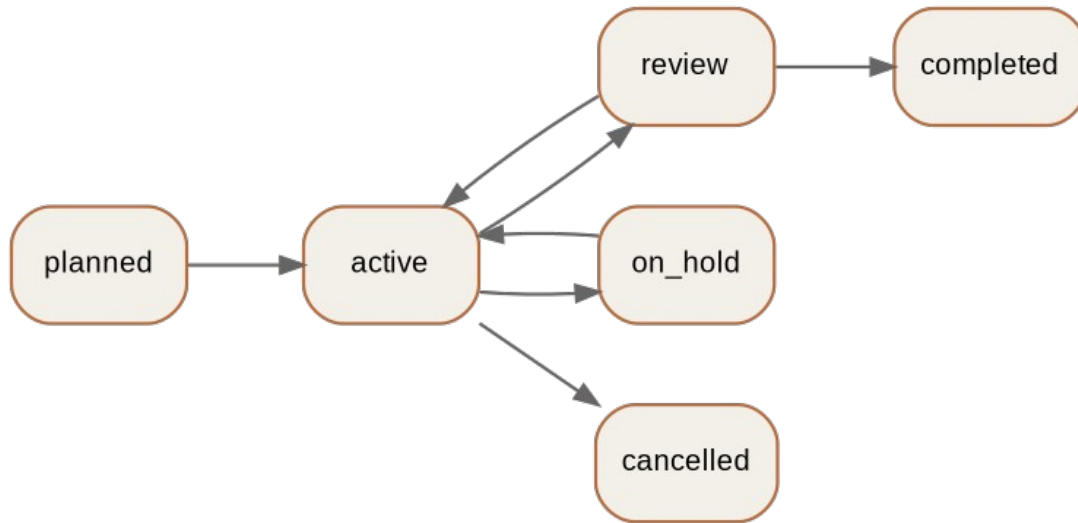


Figure 3. Default business project lifecycle.

Business Project Requirements

Area	Requirements
Project record	Client, originating lead/quotation, title, scope summary, owner, status, start/end dates, budget/contract references.
Milestones	Title, description, target date, status, sort order, client-visible flag.
Tasks	Project/milestone, assignee, priority, status, due date, estimate optional, internal/client visibility.
Files	Upload/link files with visibility controls: internal, client, public/portfolio candidate.
Approvals	Request client approval for specified deliverable/milestone; capture decision, actor, timestamp and note.
Feedback	Client-visible feedback threads separate from internal notes.
Progress	Derived from milestone/task state or explicitly managed with documented rule; avoid fake precision.
Portfolio promotion	Completed internal project may be linked to a public case study, but public content is a separate editorial entity to protect privacy.

Delivery stages align to LOGIKAiN process: Discover, Analyze, Design, Build, Test, Launch and Improve; the milestone/task model must support that workflow without hard-coding a single project methodology.

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Quotation & Sales Workflow

Sales data turns qualified opportunities into formal commercial offers and executable work.

Quotation Lifecycle

```
draft -> sent -> viewed -> accepted
      \-> rejected
      \-> revision_requested -> draft
```

Requirements

Capability	Requirement
Builder	Create quotation from lead/client with line items, quantity, unit, unit price, discount, tax configuration and notes/terms.
Numbering	Unique human-readable quotation number generated server-side.
PDF/Share	Generate stable PDF/document representation and a secure client-facing view when sent.
Status	draft, sent, viewed, accepted, rejected, expired, revision_requested, cancelled.
Acceptance	Client acceptance records actor, timestamp, accepted terms/version; do not silently mutate accepted quotation.
Revision	New revision/version preserves prior commercial history.
Conversion	Accepted quotation can create/link business project and billing schedule/invoice without duplicate data entry.

COMMERCIAL INTEGRITY

Accepted quotations and issued invoices are business records. Changes that alter totals or terms should create a new revision/credit/void workflow rather than overwrite history.

Client Portal

Clients receive controlled visibility without access to internal operational data.

Portal Modules

Module	Client capability
Dashboard	See active projects, outstanding approvals, recent files, open support and invoice status summary.
Project	View approved project summary, progress, milestones and client-visible activities.
Files	Download client-visible project/document assets; upload only where explicitly allowed.
Approvals	Review requested deliverable, approve/reject/request changes with comment.
Feedback	Submit project feedback visible to authorized internal members.
Quotations	View sent quotation versions and accept/reject/request revision when active.
Invoices	View issued invoices, due dates, payment status and downloadable invoice documents.
Support	Create and respond to support tickets for authorized client/project context.
Profile/Members	Owner-approved management of client portal contacts/members if enabled.

Portal Security

- Every portal query scopes by authenticated user -> client_members -> client/project relationship.
- Guessing a UUID/slug must not reveal another client record.
- Internal notes, margin/cost data, internal tasks and unpublished case-study content are excluded unless explicitly marked client-visible.
- Portal pages are noindex and excluded from public sitemap.
- Sensitive document URLs use access-controlled storage or short-lived signed URLs as appropriate.

Billing, Invoicing & Payments

Commercial records support project, retainer and subscription-ready billing without becoming a full accounting system.

Invoice Requirements

Capability	Requirement
Invoice record	Unique number, client, optional project, issue date, due date, status, currency, line items, subtotal, discount, tax, total, notes.
Statuses	draft, issued, partially_paid, paid, overdue, void, cancelled.
Document	Generate/download invoice PDF with immutable snapshot/version for issued state.
Payment record	Amount, currency, provider/method, provider reference, timestamp, status, metadata; can support partial payments.
Gateway	Payment provider is an adapter; webhook signatures validated; events idempotent.
Manual payment	Authorized finance user can record offline/manual payment with audit trail.
Retainer	Recurring service agreement can define billing cadence and create invoices through scheduled jobs.
Subscription-ready	Proprietary SaaS products may map product subscriptions to billing provider; do not conflate internal service retainers with SaaS tenant subscriptions.

Financial Boundaries

- This module tracks sales invoices and payment status; it is not a general ledger or tax-accounting system.
- Money values use integer minor units or exact decimal numeric types, never floating-point arithmetic.
- Issued invoice totals cannot be silently edited; use void/reissue or adjustment workflow with audit history.
- Payment webhook processing must be idempotent and resilient to retries/out-of-order events.

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Support & After-Sales System

After-sales support continues the relationship after project launch.

Ticket Model

Field/feature	Requirement
Context	Client required; project optional; category/service optional.
Priority	low, normal, high, urgent with clear operational meaning.
Status	open, in_progress, waiting_client, resolved, closed.
Assignment	Support owner/team member.
Conversation	Chronological messages with author role and attachments.
Visibility	Client sees client-visible messages only; internal notes are separate.
Resolution	Resolved timestamp, resolution summary, reopen capability.
SLA readiness	Store first_response_at/resolved_at so SLA reporting can be added without schema rewrite.

Automation & AI Layer

Automation removes repetitive work; AI assists decisions/content but does not bypass human responsibility.

Automation Jobs

Automation	Default trigger / result
Lead acknowledgment	New valid lead -> optional acknowledgement email + internal notification.
Lead assignment notification	Assignee changes -> notify responsible user.
Quotation sent	Sent status -> client notification + timeline event.
Approval request	New approval -> portal/email notification.
Invoice due reminder	Scheduled relative to due date; avoid duplicate sends.
Support update	Ticket/message state change -> relevant notifications.
Content publish	Publish -> revalidate affected routes/sitemap and optionally notify internal channel.
SEO monitoring	Scheduled ingestion/check of Search Console or configured reporting data when integration enabled.

AI-Assisted Capabilities

- Lead qualification suggestion based on submitted brief; human owns final status.
- Project/meeting summary assistance from user-provided data where legally appropriate.
- SEO metadata suggestions and content-quality checklist; never guarantee ranking.
- Article/project case-study drafting assistance using existing factual records; human review before publish.
- Support response drafting; human can edit before sending.
- Provider abstraction supports OpenAI/Gemini or another approved model without embedding credentials in client code.

AI SAFETY & PROVENANCE

AI output that could change public claims, prices, commitments, client decisions or support responses must be reviewable before external publication/sending. Do not invent project results, client names, metrics or testimonials.

SaaS / Proprietary Product Foundation

The corporate platform must be capable of becoming an ecosystem without prematurely forcing every internal module into multi-tenancy.

Separation Principle

LOGIKaIn internal operations use clients and internal business projects. Future proprietary products use a distinct product tenant boundary so a SaaS customer account is not automatically treated as a consulting client record. A link may exist when the same organization uses both.

Foundation Entities

Entity	Purpose
products	Catalog of proprietary LOGIKaIn products/tools.
product_tenants	Customer tenant/workspace for a product.
product_tenant_members	User membership and product role.
product_plans	Plan definition and feature limits.
product_subscriptions	Tenant-plan subscription state and provider references.
product_usage_events	Metering/usage events when a product needs them.
product_entitlements	Resolved or explicit feature entitlements when plan logic requires.

Multi-Tenant Rules

- Every tenant-scoped table includes `tenant_id` and RLS isolates tenants.
- Cross-tenant administration is owner/support-only and explicitly audited.
- Product subscription/payment logic uses provider adapters and webhook idempotency.
- Do not migrate CMS/CRM tables into tenant scope just to reuse the word "tenant"; keep bounded contexts clear.

Media, Search & Notification Services

Shared services support every major module without duplicating implementation.

Media Library

- Store asset metadata separately from object storage path; include mime type, size, dimensions when known, alt text, owner/uploader and visibility classification.
- Public marketing media may be public/CDN-backed; private client/project/finance assets use protected buckets/policies or signed access.
- Deleting an asset referenced by content or a business record must be blocked or require explicit replacement workflow.
- Image optimization/responsive sizing must prevent original high-resolution assets from being loaded into small listing cards.

Search

Search surface	Behavior
Public projects	Keyword + service + industry + technology filters; stable pagination.
Public insights	Keyword/category filters; avoid indexable explosion of arbitrary query URLs.
Admin content	Server-side/paginated search by title/slug/status.
CRM/clients	Search name/company/email/phone where authorized; index sensitive fields appropriately.
Projects/tasks	Search/filter by project/status/assignee/due date.
Support	Search ticket subject/reference/status/client.

Notifications

- In-app notification record with type, recipient, entity reference, read state and timestamps.
- Email delivery is secondary transport and must not be the only record of critical state.
- Notification preferences may be added per user/channel; security-critical notifications can remain mandatory.

Security, Auth, RLS & Privacy

Security boundaries are enforced at authentication, server authorization, database policy and storage access layers.

Security Model

Layer	Required control
Authentication	Supabase Auth; secure session handling; email/password or approved provider.
Authorization	Capability checks in server code plus PostgreSQL RLS for exposed tables.
Database	RLS enabled for every exposed schema table containing protected data. Policies explicitly cover SELECT/INSERT/UPDATE/DELETE as needed.
Storage	Bucket/object policies separate public marketing assets from protected client/internal files.
Secrets	Service-role keys, payment secrets, AI keys and webhook secrets remain server-only environment variables.
Input	Server validation for every mutation and external webhook payload.
CSRF/Origin	Use framework-safe mutation patterns and validate origin/cookies where applicable.
Rate/abuse	Rate limit public forms/auth-sensitive endpoints; anti-spam controls on inbound forms.
Audit	Privilege, billing, approval, publishing and destructive actions logged.
Privacy	Collect only needed personal/business data; define retention/export/deletion processes when legally/operationally required.

RLS Policy Intent

Data class	Anonymous	Authenticated internal	Client member
Published content	SELECT published only	Role-based CRUD	SELECT public only
Leads/internal notes	INSERT public lead form via controlled server path; no SELECT	Authorized sales/admin	None
Clients/projects	None	Authorized internal	Own client/project subset only
Invoices/payments	None	Finance/admin/owner	Own issued invoices/payment view only
Support	None	Support/admin/project authorized	Own client tickets/messages
Activity logs	None	Admin/owner; scoped internal as needed	None

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Data Model & ERD

The relational model is divided into bounded contexts that share stable identifiers and explicit relationships.

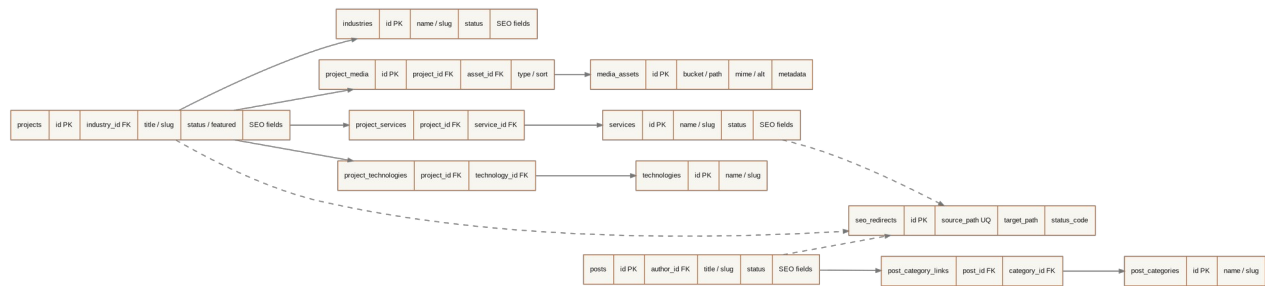


Figure 4. Content, portfolio and SEO ERD.

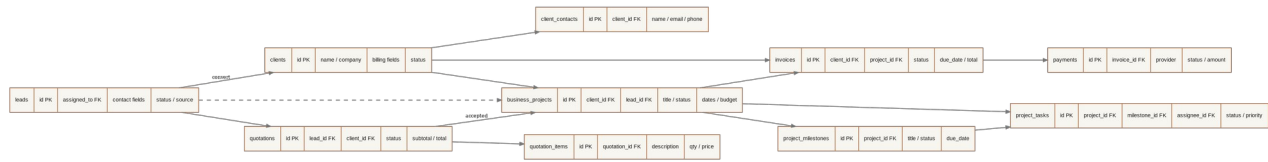


Figure 5. CRM, client, sales, delivery and billing ERD.

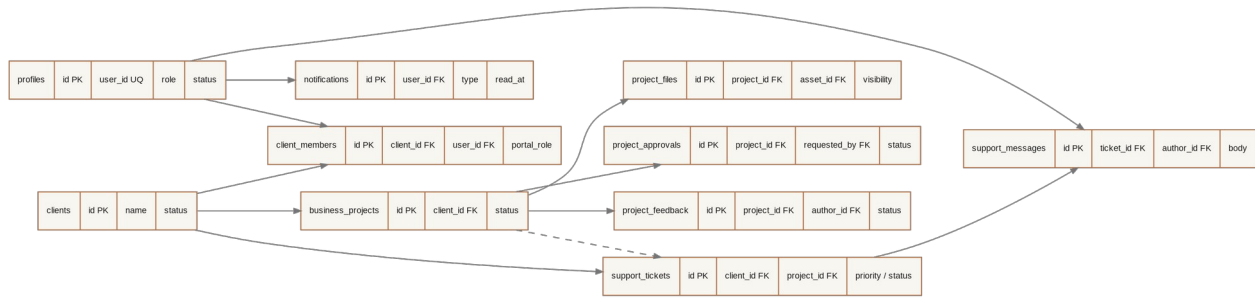


Figure 6. Portal, project collaboration and support ERD.

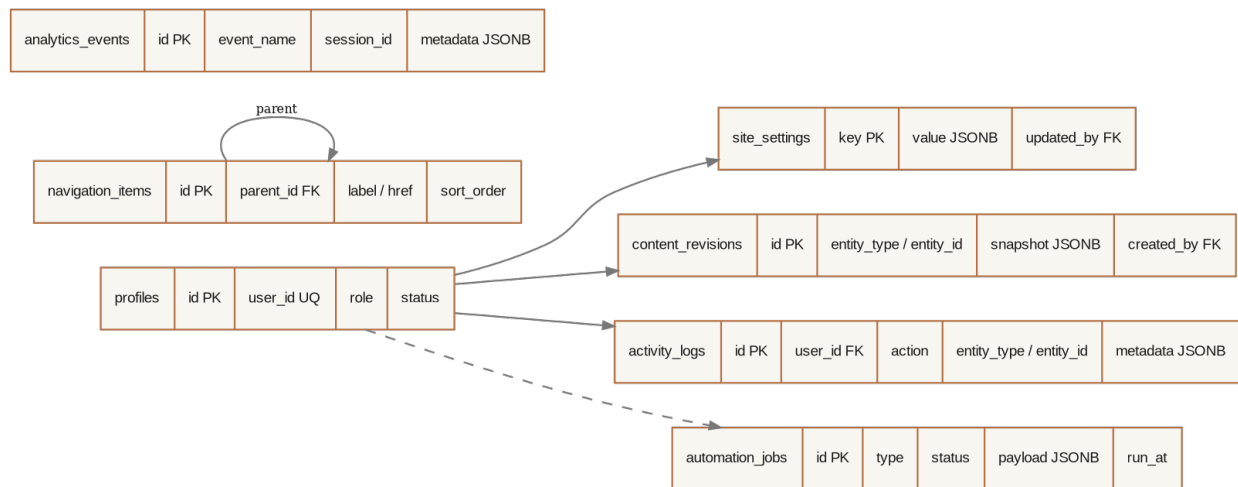


Figure 7. System, audit, settings, automation and analytics ERD.

ERD RULE

Diagrams show the conceptual core. The database catalog in the next section is authoritative for table responsibilities. AI agents must preserve foreign-key integrity and use migrations for any additional implementation fields.

Database Schema Specification

Authoritative table responsibilities for the final platform. Exact SQL types may be refined without changing entity meaning.

Identity & System

Table	Core fields / responsibility
profiles	id PK; user_id UQ FK auth.users; full_name; avatar_url; role; status; timestamps
site_settings	key PK; value JSONB; updated_by FK; updated_at
navigation_items	id PK; parent_id self FK; location; label; href; sort_order; status
activity_logs	id PK; user_id FK; action; entity_type; entity_id; metadata JSONB; created_at
content_revisions	id PK; entity_type; entity_id; version; snapshot JSONB; created_by FK; created_at
notifications	id PK; user_id FK; type; title; body; entity_type/id; read_at; created_at
automation_jobs	id PK; type; status; payload JSONB; run_at; attempts; last_error; idempotency_key UQ

Public Content & SEO

Table	Core fields / responsibility
projects	public case-study entity; id; title; slug UQ; narrative fields; industry_id FK; status; featured; SEO; timestamps
project_media	id; project_id FK; asset_id FK; media_type; caption; sort_order
services	id; name; slug UQ; descriptions; icon; status; featured; sort_order; SEO
industries	id; name; slug UQ; description; status; SEO
technologies	id; name; slug UQ; icon_url; website_url
project_services	project_id + service_id composite PK
project_technologies	project_id + technology_id composite PK
posts	id; author_id FK; title; slug UQ; excerpt; content; status; published_at; SEO
post_categories	id; name; slug UQ
post_category_links	post_id + category_id composite PK
testimonials	id; client display fields; content; published; sort_order
media_assets	id; bucket; path; mime; size; width/height; alt_text; visibility; uploaded_by; timestamps
seo_redirects	id; source_path UQ; target_path; status_code; reason; created_by; created_at

CRM & Client

Table	Core fields / responsibility
leads	id; identity/contact fields; company; service_interest; budget_range; timeline; description; source; status; assigned_to; timestamps
lead_notes	id; lead_id FK; author_id FK; body; created_at

Table	Core fields / responsibility
clients	id; name/company; legal/billing fields; status; originating_lead_id optional; timestamps
client_contacts	id; client_id FK; name; email; phone; role/title; primary flag
client_members	id; client_id FK; user_id FK; portal_role; status; UQ client+user
client_notes	id; client_id FK; author_id; body; created_at

Delivery & Collaboration

Table	Core fields / responsibility
business_projects	id; client_id FK; lead_id optional; quotation_id optional; title; scope; owner_id; status; dates; budget fields; timestamps
project_milestones	id; project_id FK; title; description; status; target_date; client_visible; sort_order
project_tasks	id; project_id FK; milestone_id optional; assignee_id; title; description; status; priority; due_date; client_visible
project_files	id; project_id FK; asset_id FK; visibility; label; uploaded_by; created_at
project_approvals	id; project_id FK; milestone/task optional; requested_by; client_member/user actor; status; request_note; decision_note; timestamps
project_feedback	id; project_id FK; author_user_id; body; status; created_at

Sales & Billing

Table	Core fields / responsibility
quotations	id; quote_number UQ; lead_id optional; client_id optional; status; currency; subtotal; discount; tax; total; valid_until; version; timestamps
quotation_items	id; quotation_id FK; service_id optional; description; qty; unit; unit_price; discount; tax; line_total; sort_order
invoices	id; invoice_number UQ; client_id FK; project_id optional; status; currency; issue_date; due_date; totals; version; timestamps
invoice_items	id; invoice_id FK; description; qty; unit; unit_price; tax; line_total; sort_order
payments	id; invoice_id FK; provider; provider_reference; method; amount; currency; status; paid_at; metadata JSONB
service_agreements	id; client_id FK; project_id optional; type retainer/maintenance; billing cadence; status; start/end; amount/currency

Support

Table	Core fields / responsibility
support_tickets	id; reference UQ; client_id FK; project_id optional; subject; category; priority; status; assigned_to; first_response_at; resolved_at; timestamps
support_messages	id; ticket_id FK; author_user_id; body; visibility; created_at
support_attachments	id; message_id FK; asset_id FK

SaaS Foundation

Table	Core fields / responsibility
products	id; name; slug UQ; status; metadata JSONB
product_tenants	id; product_id FK; name; slug/key; status; linked_client_id optional
product_tenant_members	id; tenant_id FK; user_id FK; role; status; UQ tenant+user
product_plans	id; product_id FK; code; name; price metadata; limits JSONB; status
product_subscriptions	id; tenant_id FK; plan_id FK; provider; provider_subscription_id; status; period dates; metadata
product_usage_events	id; tenant_id FK; event_type; quantity; occurred_at; metadata
product_entitlements	id; tenant_id FK; feature_key; value JSONB; source

API / Server Contract & Integration Boundaries

The product may use Server Actions and route handlers internally, but contracts remain explicit and testable.

Server Operation Categories

Category	Examples	Rules
Public reads	getPublishedProjects, getServiceBySlug, getPostBySlug	Published-only; cacheable/revalidated; never return draft/private fields.
Public writes	submitContact, submitProjectRequest	Validated, rate-limited/anti-spam, minimal permissions, create lead + event.
CMS writes	create/update/publish content, media operations	Authenticated capability + RLS + revision/activity event.
CRM writes	status/note/assign/convert	Authorized sales/admin; preserve history.
Project writes	tasks, milestones, files, approvals	Project authorization; visibility controls.
Portal writes	approval decision, feedback, support message	Authenticated + client membership + entity scope.
Billing	issue invoice, record payment, webhook ingestion	Finance/admin; immutable issued snapshots; idempotent webhook.
Automation	enqueue/process job	Server-only; idempotency key; retry policy and error visibility.

Integration Interfaces

Interface	Contract expectation
EmailProvider	sendTransactional(template, recipient, variables); failure/retry result; provider message id.
PaymentProvider	createPayment/checkout where used, verifyWebhook, normalizeEvent, refund/void hooks if enabled.
AIProvider	generate structured result with model/provider metadata; server-only secret; timeout/retry; moderation/policy hook as needed.
AnalyticsAdapter	track server/client events without coupling business logic to vendor API.
StorageService	public/private upload, signed URL, metadata validation and deletion safeguards.

Analytics, Observability & Auditability

Measure acquisition, product usage and operational health without leaking sensitive data.

Business / Product Events

Event	Purpose
page_view	Traffic and popular routes.
service_view / project_view / article_view	Content demand and interest.
cta_click	Conversion intent.
contact_submit / project_request_submit	Inbound conversion.
lead_status_changed	Pipeline movement.
quotation_sent / accepted	Sales progression.
portal_login / approval_decided	Client engagement.
invoice_issued / payment_recorded	Commercial lifecycle.
support_ticket_created / resolved	After-sales operations.

Observability

- Capture application/server errors with request context but never log secrets, auth tokens, payment secrets or unnecessary personal data.
- Background job failures must be queryable with attempts and last_error; critical automation must not fail silently.
- Audit logs are immutable product history; they are not a substitute for error monitoring.
- Analytics adapters are isolated; the platform remains functional if analytics is unavailable.

Performance, Accessibility & Quality Budgets

Performance and accessibility are release requirements, not optional polish.

Performance Budget

Area	Requirement
Rendering	Server Components by default on public pages; Client Components only for interaction.
JavaScript	Avoid heavy global client bundles; code-split interactive tools; admin complexity must not leak into public bundle.
Images	Responsive sizes, modern formats where appropriate, dimensions reserved to prevent layout shift, lazy-load below fold.
Fonts	Limit families/weights; preload only critical font assets; use stable fallbacks.
Third-party scripts	Load only justified scripts; defer/consent-gate where appropriate.
Database	Index frequent filters/foreign keys; paginate large lists; avoid N+1 query patterns.
Caching	Use route/data caching and explicit revalidation after CMS publishes/updates.

Accessibility

- Semantic heading hierarchy and landmarks; one meaningful H1 per page context.
- Keyboard-operable navigation, dialogs, forms, tables and interactive controls with visible focus.
- Sufficient contrast using approved brand colors; never rely on color alone for state.
- Form labels/errors are programmatically associated and announced appropriately.
- Meaningful images require alt text; decorative imagery is omitted from accessibility tree.
- Respect reduced-motion; no essential content available only through hover.

Testing Strategy & Release Gates

Every phase has automated and manual gates before the next dependent phase is considered complete.

Test Layers

Layer	Coverage
Unit	Validation, money calculations, permission helpers, SEO/canonical helpers, state transitions.
Database / RLS	Policy tests for anonymous/internal/client roles; constraints, foreign keys and migration behavior.
Integration	Lead creation, quotation conversion, project relations, invoice/payment flows, provider adapters.
E2E	Public navigation/forms, admin auth/publishing, sales flow, client portal scope, approval/support flows.
SEO smoke	Status, title, description, canonical, robots meta, H1, structured data presence, sitemap inclusion/exclusion, redirects.
Accessibility	Automated checks + keyboard/manual verification of critical paths.
Performance	Lighthouse/lab regression on representative public routes; field monitoring post-launch.
Security	Privilege escalation attempts, object ID guessing, storage policy, secret exposure, webhook signature/idempotency.

Release Blocking Defects

- Unauthorized data exposure or broken RLS.
- Lead/contact form loses valid submissions without visible failure.
- Published route returns draft/private content.
- Canonical/sitemap/index rules create broad accidental deindexation or duplicate-index explosion.
- Quotation/invoice totals use incorrect arithmetic or issued commercial records can be silently overwritten.
- Client can access another client/project by changing URL identifier.
- Deployment lacks migration/rollback/backup plan.

Environments, Deployment, Backup & Recovery

Production operations must be reproducible and recoverable.

Environments

Environment	Purpose	Data rule
Local	Developer/agent implementation and tests.	Seed/synthetic data only.
Preview/Staging	PR/feature QA, integration tests, stakeholder review.	No uncontrolled production PII copy.
Production	Official website and operational system.	Protected, backed up, monitored.

Deployment Requirements

- Environment schema validates required variables at startup/build where appropriate.
- Database migrations run through controlled process before incompatible application changes are activated.
- Production deployment includes smoke tests for home, representative service/project/article, contact submission path, admin login and portal authorization.
- Backup policy covers PostgreSQL data and critical private assets/document snapshots according to available provider capabilities and business needs.
- Rollback procedure documents application rollback plus database migration safety; destructive migrations require expand/migrate/contract strategy rather than single irreversible step.

Master Development Roadmap - Phase 1 to 17

One product. One final scope. Phases define dependency order and completion gates, not separate product versions.

Phase	Required deliverables
Phase 1 - Foundation	Repository, Next.js/TypeScript, design tokens, environment schema, Supabase project, migrations, base layouts, canonical host config.
Phase 2 - Database & Security	Core identity/content/business tables, Auth, RLS, storage policies, seed data, typed data access, policy tests.
Phase 3 - SEO Engine	Metadata/canonical helpers, robots/sitemap, structured-data helpers, redirects, breadcrumbs, revalidation, crawler tests.
Phase 4 - Public Company Website	Home, About, Services, Industries, Process, Projects, Case Studies, Insights, Contact, Start Project, legal.
Phase 5 - CMS	Dashboard, content CRUD, media, homepage/navigation/settings, users, activity, revisions, SEO editor.
Phase 6 - Content & Portfolio Engine	Publishing workflow, case-study builder, metrics/media, relationships, preview, revision restore.
Phase 7 - SEO Content Engine	Topic clusters, service/industry landing pages, internal linking, SERP preview, Search Console/analytics hooks.
Phase 8 - CRM & Lead Management	Lead capture, pipeline, notes, assignment, dedupe, search/filter, conversion readiness.
Phase 9 - Client Management	Clients, contacts, memberships, relationship timeline and linked commercial/project records.
Phase 10 - Project Management	Business projects, milestones, tasks, files, approvals, feedback, progress and delivery activity.
Phase 11 - Quotation & Sales	Quotation builder, versions, share/PDF, status, acceptance, lead-to-client/project conversion.
Phase 12 - Client Portal	Authenticated dashboard, projects, files, approvals, quotations, invoices and support.
Phase 13 - Billing	Invoices, items, payment records, due status, manual/provider payment, retainer/subscription-ready abstractions.
Phase 14 - Support System	Tickets, messages, attachments, priority/status, assignment, resolution history.
Phase 15 - Automation & AI	Background jobs, notifications, reminders, AI-assisted content/lead/project/support workflows with human review.
Phase 16 - SaaS / LOGIKAIin Products	Products, tenants, members, plans, subscriptions, usage/entitlements and tenant isolation.
Phase 17 - Hardening & Scale	CWV/performance, accessibility, security audit, SEO crawler tests, E2E, backups, monitoring, final production readiness.

Roadmap Execution Rule

SEQUENTIAL BUT INTEGRATED

An AI agent may implement multiple phases in one working session, but it must respect dependencies and phase gates. It must not call a downstream module complete if upstream data, auth, policies, migrations or tests are placeholders. All phase deliverables belong to this same final PRD.

Functional Acceptance Criteria

The platform is complete only when the following end-to-end behaviors pass.

ID	Acceptance criterion
AC-001	Create a project case study in CMS, preview it, publish it and see it appear on index/detail routes without redeployment.
AC-002	Published service/industry/project/post has valid metadata, canonical, sitemap inclusion and server-rendered primary content.
AC-003	Changing published slug creates a working permanent redirect and removes old URL from canonical sitemap.
AC-004	Anonymous user cannot read any draft, lead, client, project-operation, invoice, support or activity-log record.
AC-005	Valid start-project form creates exactly one lead per request and records source/status safely.
AC-006	Sales user can qualify lead, create quotation and convert accepted opportunity to client + project with linked history.
AC-007	Quotation version accepted by client cannot be silently overwritten by later edits.
AC-008	Internal project supports milestones, tasks, files and client approval request.
AC-009	Portal member sees only records for their client membership and cannot access another client by URL manipulation.
AC-010	Client approval decision records actor, timestamp and note and appears in internal project timeline.
AC-011	Finance can issue invoice from client/project, record partial/full payment and derive correct status without floating-point errors.
AC-012	Payment webhook retries do not create duplicate payments.
AC-013	Client can create support ticket and exchange client-visible messages; internal notes remain private.
AC-014	Background reminder/notification jobs are idempotent and failed jobs are visible for retry/debugging.
AC-015	Owner can inspect activity logs for publishing, privilege, quotation/invoice, approval and destructive actions.
AC-016	High-value content revision can be restored without losing current audit history.
AC-017	Production deployment can be reproduced using documented environment, migration, seed/bootstrap and smoke-test process.
AC-018	Representative public routes pass accessibility keyboard checks and meet performance regression budgets.
AC-019	SaaS tenant policy test proves tenant A cannot read tenant B records.
AC-020	No critical feature depends on placeholder data, fake metrics or disabled security policies.

Non-Functional Requirements

Cross-cutting standards apply to every module.

Category	Requirement
Security	Least privilege, RLS, secret isolation, validated server mutations, secure storage, signed webhooks, audit critical actions.
Performance	Fast public pages, bounded client JS, pagination, indexed queries, optimized media, revalidation.
Reliability	Idempotent external events/jobs, safe retries, visible failures, transactional multi-record conversions.
Maintainability	TypeScript strictness, module boundaries, migrations, shared validation, documented provider interfaces.
Scalability	Pagination, background jobs, tenant-aware product tables, no per-record static code.
Accessibility	WCAG-aligned semantic/keyboard/contrast/form behavior on public, admin and portal critical flows.
SEO	Server-visible content, canonical/index discipline, sitemap/robots/redirect integrity and structured data tied to actual content.
Privacy	Minimize collection, control access, avoid logging sensitive fields, define retention/export/deletion handling.
Brand consistency	Charcoal/Copper/Cream design tokens, typography system, consistent components; no generic AI-template aesthetic.
Auditability	State-changing business records and privilege/publishing actions attributable to actor/time.

AI Agent Handoff & Definition of Done

Instructions for an autonomous coding agent implementing this PRD.

Non-Negotiable Agent Rules

8. Read this document as the source of truth. Do not regress to V1/V2/V3 scope labels; phases are implementation sequence only.
9. Build production-capable code, not disconnected mock screens. UI is incomplete until it reads/writes the specified data with correct authorization.
10. Create migrations and RLS policies alongside tables. Never temporarily disable RLS in production-oriented implementation to make a feature work.
11. Use server-first rendering for public content. Do not convert the marketing site into a client-only SPA.
12. Do not ship placeholder buttons, fake counters, fake testimonials, invented client names/results or nonfunctional menu items as completed features.
13. Use adapters for third-party providers when credentials/vendor decisions are not supplied; implement a safe local/mock adapter only for testing with obvious non-production status.
14. Preserve commercial record history. Do not overwrite accepted quotations, issued invoices or payment events in place.
15. Every phase completion requires tests and integration with previous phases; update documentation when architecture/contracts change.
16. Run lint/typecheck/tests/build before declaring completion, and resolve errors rather than suppressing them broadly.
17. At final handoff provide environment-variable inventory, migration instructions, seed/bootstrap instructions, deployment steps, admin bootstrap method, backup/restore guidance and known limitations.

Definition of Done

Dimension	Done when
Functionality	All Must requirements and acceptance criteria relevant to implemented phases pass end-to-end.
Data	Schema, constraints, indexes, migrations and seeds/bootstrap are committed and reproducible.
Security	RLS/policies and server authorization tested for anonymous, internal roles and client members.
SEO	Published pages render correct status/metadata/canonical/schema/sitemap behavior.
UI/UX	Responsive, accessible, loading/error/empty/success states implemented; brand design consistent.
Tests	Unit/integration/E2E/SEO/security tests cover critical flows and pass.
Operations	Deployment, environment, backups, monitoring and rollback documented.
Documentation	README/architecture notes reflect actual implementation; no hidden manual setup required beyond documented steps.

FINAL INSTRUCTION TO AI AGENT

Do not stop at visually complete. A module is complete only when UI, server logic, database, authorization, workflow, tests and operational behavior work together as specified.

APPENDIX A

Route Inventory

Canonical route families for public, admin and portal surfaces.

Route	Surface	Index / Access
/	Public	Index
/about	Public	Index
/services	Public	Index
/services/[slug]	Public	Index published
/industries	Public	Index
/industries/[slug]	Public	Index published
/projects	Public	Index
/projects/[slug]	Public	Index published
/insights	Public	Index
/insights/[slug]	Public	Index published
/process	Public	Index
/contact	Public	Index
/start-project	Public	Index
/privacy	Public	Index
/terms	Public	Index
/login	Auth	Noindex
/admin	Admin	Noindex + auth
/admin/projects	Admin	Noindex + auth
/admin/services	Admin	Noindex + auth
/admin/industries	Admin	Noindex + auth
/admin/insights	Admin	Noindex + auth
/admin/media	Admin	Noindex + auth
/admin/leads	Admin	Noindex + auth
/admin/clients	Admin	Noindex + auth
/admin/delivery/projects	Admin	Noindex + auth
/admin/quotations	Admin	Noindex + auth
/admin/invoices	Admin	Noindex + auth
/admin/support	Admin	Noindex + auth
/admin/seo	Admin	Noindex + auth
/admin/automation	Admin	Noindex + auth
/admin/users	Admin	Noindex + auth
/admin/activity	Admin	Noindex + auth
/admin/settings	Admin	Noindex + auth
/portal	Client portal	Noindex + auth + membership

Route	Surface	Index / Access
/portal/projects	Client portal	Noindex + auth + membership
/portal/projects/[id]	Client portal	Noindex + scoped membership
/portal/files	Client portal	Noindex + scoped membership
/portal/approvals	Client portal	Noindex + scoped membership
/portal/quotations	Client portal	Noindex + scoped membership
/portal/invoices	Client portal	Noindex + scoped membership
/portal/support	Client portal	Noindex + scoped membership

APPENDIX B

Status Enums & State Machines

Controlled values are explicit to prevent incompatible strings across UI/database/business logic.

Enum	Allowed values
content_status	draft, review, published, archived
lead_status	new, contacted, qualified, proposal, negotiation, won, lost, spam
client_status	active, inactive, archived
business_project_status	planned, active, on_hold, review, completed, cancelled
milestone_status	planned, in_progress, blocked, review, completed, cancelled
task_status	todo, in_progress, blocked, review, done, cancelled
priority	low, normal, high, urgent
quotation_status	draft, sent, viewed, accepted, rejected, expired, revision_requested, cancelled
invoice_status	draft, issued, partially_paid, paid, overdue, void, cancelled
payment_status	pending, succeeded, failed, refunded, cancelled
approval_status	pending, approved, rejected, changes_requested, cancelled
ticket_status	open, in_progress, waiting_client, resolved, closed
job_status	queued, processing, succeeded, failed, cancelled
portal_role	client_owner, client_admin, client_member, viewer
asset_visibility	public, internal, client
product_subscription_status	trialing, active, past_due, paused, cancelled, expired

APPENDIX C

Database Table Catalog

Condensed list for migration planning and implementation sequencing.

Domain	Table	Primary responsibility
Identity & System	profiles	id PK
Identity & System	site_settings	key PK
Identity & System	navigation_items	id PK
Identity & System	activity_logs	id PK
Identity & System	content_revisions	id PK
Identity & System	notifications	id PK
Identity & System	automation_jobs	id PK
Public Content & SEO	projects	public case-study entity: id
Public Content & SEO	project_media	id
Public Content & SEO	services	id
Public Content & SEO	industries	id
Public Content & SEO	technologies	id
Public Content & SEO	project_services	project_id + service_id composite PK
Public Content & SEO	project_technologies	project_id + technology_id composite PK
Public Content & SEO	posts	id
Public Content & SEO	post_categories	id
Public Content & SEO	post_category_links	post_id + category_id composite PK
Public Content & SEO	testimonials	id
Public Content & SEO	media_assets	id
Public Content & SEO	seo_redirects	id
CRM & Client	leads	id
CRM & Client	lead_notes	id
CRM & Client	clients	id
CRM & Client	client_contacts	id
CRM & Client	client_members	id
CRM & Client	client_notes	id
Delivery & Collaboration	business_projects	id
Delivery & Collaboration	project_milestones	id
Delivery & Collaboration	project_tasks	id
Delivery & Collaboration	project_files	id
Delivery & Collaboration	project_approvals	id
Delivery & Collaboration	project_feedback	id
Sales & Billing	quotations	id
Sales & Billing	quotation_items	id
Sales & Billing	invoices	id

Domain	Table	Primary responsibility
Sales & Billing	invoice_items	id
Sales & Billing	payments	id
Sales & Billing	service_agreements	id
Support	support_tickets	id
Support	support_messages	id
Support	support_attachments	id
SaaS Foundation	products	id
SaaS Foundation	product_tenants	id
SaaS Foundation	product_tenant_members	id
SaaS Foundation	product_plans	id
SaaS Foundation	product_subscriptions	id
SaaS Foundation	product_usage_events	id
SaaS Foundation	product_entitlements	id

APPENDIX D

SEO Route / Rendering Matrix

Default behavior for public and protected route families.

Route family	Rendering	Indexing	Canonical	Structured data	Metadata source
Home / About / Process	Static/cached server	Index	Self canonical	Organization/WebPage as applicable	Manual + settings
Service detail	Static/revalidated server	Published only	Self canonical	Service + BreadcrumbList	CMS-driven
Industry detail	Static/revalidated server	Published only	Self canonical	WebPage + BreadcrumbList	CMS-driven
Project detail	Static/revalidated server	Published only	Self canonical	CreativeWork/WebPage + BreadcrumbList as applicable	CMS-driven
Article detail	Static/revalidated server	Published only	Self canonical	Article + BreadcrumbList	CMS-driven
Indexes	Server/cached + pagination	Index canonical pages	Stable pagination canonical	CollectionPage/ Breadcrumb optional	System + CMS
Filter/search query URLs	Dynamic	Noindex or canonicalized by policy	To stable intended page	None unless justified	System
Preview	Dynamic protected/signed	Noindex	None/self preview	None	CMS
Admin / Portal	Dynamic authenticated	Noindex	N/A	None	System

APPENDIX E

Official Technical References

Primary references used for architecture requirements; implementation should re-check current supported versions/security patches at build time.

Reference	URL / relevance
LOGIKAI Company Profile Advanced v4 (2026)	Internal source document: positioning, services, markets, process, brand and long-term direction.
Next.js - Metadata Files: robots.txt	https://nextjs.org/docs/app/api-reference/file-conventions/metadata/robots
Next.js - Metadata Files: sitemap.xml	https://nextjs.org/docs/app/api-reference/file-conventions/metadata/sitemap
Google Search Central - JavaScript SEO basics	https://developers.google.com/search/docs/crawling-indexing/javascript/javascript-seo-basics
Google Search Central - Dynamic rendering guidance	https://developers.google.com/search/docs/crawling-indexing/javascript/dynamic-rendering
Google Search Central - Core Web Vitals	https://developers.google.com/search/docs/appearance/core-web-vitals
Supabase - Row Level Security	https://supabase.com/docs/guides/database/postgres/row-level-security
Supabase - Auth	https://supabase.com/docs/guides/auth
Supabase - Storage access control	https://supabase.com/docs/guides/storage/security/access-control

VERSION POLICY

Do not pin a stale patch version in perpetuity. Use the supported Next.js 16.x/security release compatible with the repository at implementation time, and apply current security guidance before production deployment.

LOGIKain

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One product. One source of truth. Build by phase, integrate end-to-end.

Think Clearly. Build Logically.